Alzheimer's Disease as Attractor Medicine

On the Two Simultaneous Identity Drift Events in Neurodegeneration, Their Geometric Structure, Their Drug Targets, and the Difference Between *Neuronal Identity and Memory Identity*

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Alzheimer's disease is two simultaneous identity drift events, not one. The neurons are losing their neuronal identity (MEF2C/NEUROD2 falling, DYRK1A rising, REST sequestered by tau aggregates, cells attempting aberrant cell cycle re-entry). The engram cells are simultaneously losing their memory identity (CREB/BDNF/TrkB falling, synaptic H3K27ac eroding, the informational structure of specific memories becoming inaccessible). These have different mechanisms, different drug targets, and different intervention windows. The framework resolves both. The neuron can be rescued. The information may not be lost — only inaccessible.

Abstract

We apply the Identity Axis Framework to Alzheimer's disease, identifying it as the first framework entry with two simultaneous and mechanistically distinct identity drift events operating at two different organisational levels.

Drift Event 1 (Cellular): Cortical and hippocampal neurons lose their committed neuronal identity. The Identity Anchor is MEF2C / NEUROD2; the Convergence Hub is DYRK1A. DYRK1A phosphorylates and inactivates MEF2C (Identity Anchor suppression) while simultaneously phosphorylating tau (accelerating aggregation) and promoting aberrant cell cycle re-entry. REST, which normally acts as an epigenetic guardian of the correct neuronal attractor in aging, is sequestered in autophagosomes by tau aggregates, removing the checkpoint. The identity axis ratio is $R_{\mathrm{AD}} = \mathrm{MEF2C/DYRK1A}$; the drug target derived from the Hub is a CNS-penetrant DYRK1A inhibitor. Leucettine L41 and GNF2133 are in trials.

Drift Event 2 (Memory): Engram cells lose their memory identity — the specific synaptic and epigenetic signature that encodes a particular memory. The Identity Anchor is the CREB/BDNF/TrkB axis; the Hub is HDAC2/HDAC3 overactivity and histone deacetylation of H3K27ac at memory-encoding loci. The memory is not erased but *inaccessible*: the informational structure has drifted below

retrieval threshold. The drug target is HDAC2/3 inhibition; TrkB agonism; and targeted chromatin-remodelling approaches that restore H3K27ac at engram loci without global genomic effects.

We derive the geometric sub-type structure (shallow vs. deep R_{AD}), the connection between the Down syndrome trisomy 21/DYRK1A overexpression and accelerated AD onset, the relationship between cognitive reserve and high R_{AD}, and the structural distinction between *lost* memory and *inaccessible* memory with implications for therapeutic intervention timing. Eight falsifiable predictions are stated for the record, timestamped 2026-03-09.

Keywords: Alzheimer's disease; attractor geometry; MEF2C; DYRK1A; REST; engram; memory identity; CREB; BDNF; TrkB; tau; epigenetic drift; neurodegeneration; identity drift; HDAC2; cognitive reserve; Down syndrome; OrganismCore; Identity Axis Framework

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1. Preamble: The Two Levels of Identity in Alzheimer's Disease

The author's initial observation, stated before the formal derivation, was:

> *"Alzheimer's and dementia — I believe that is absolutely a matter of identity drift of some sort. It is a person's memory structure fading."*
>
> — Eric Robert Lawson, 2026-03-09

This is correct. But more precisely than stated.

Memory structure fading captures the clinical phenomenology but conflates two mechanistically distinct events that produce the same clinical presentation by different routes:

- A **person's memory fading** because the *neuron that holds the memory has died* — this is the cellular identity drift: the neuron lost its identity, re-entered the cell cycle, and died. The memory encoded in that neuron is gone because the substrate is gone.
- A **person's memory becoming inaccessible** because the *informational structure encoding the memory has drifted* — the neuron is alive but the synaptic and epigenetic signature that constitutes the memory has eroded below retrieval threshold. The information may be recoverable.

These are the same outcome observationally. They are different mechanisms geometrically. The distinction matters for treatment: one is irreversible (dead neurons); the other may be reversible (inaccessible but not destroyed memories).

The most important clinical implication of the two-drift model: The intervention window for Drift Event 2 (memory identity) is *earlier* than the intervention window for Drift Event 1 (neuronal death). A patient in early Alzheimer's disease has neurons that are drifting but not yet dead. Their memories may be inaccessible but not yet erased. Intervention at this stage can target both drifts simultaneously. A patient in late Alzheimer's disease has lost neurons irreversibly. Drift Event 1 intervention cannot restore the dead neurons. Drift Event 2 intervention is the only remaining option, and it is limited by how much engram cell substrate survives. This is why early detection and intervention is not merely preferable — it is geometrically necessary.

2. Framework Foundations

2.1. The Identity Axis Theorem

The framework rests on five axioms [1]:

- Every committed cell occupies a stable attractor in the Waddington epigenetic landscape.
- Disease is a cell displaced into a false attractor, maintained by a convergent Hub.

- The Identity Anchor $\mathcal{A}$ is the element whose activity most directly constitutes what the cell is.
- The Convergence Hub $\mathcal{H}$ is the element whose activity most directly maintains the displaced state.
- The ratio $R = \mathcal{A}/\mathcal{H}$ is a continuous measurable coordinate; $\mathcal{H}$ is the primary drug target.

2.2. The Inverted REST Geometry: A Framework Novelty

Before the two drift events are derived, a structural novelty must be stated explicitly. In the framework's treatment of small cell lung cancer (SCLC) subtype transitions [1], REST was identified as the Convergence Hub that *silences* neuronal identity in non-neuroendocrine SCLC cells: REST rising = neuronal identity lost.

In Alzheimer's disease, the geometry is *inverted*:

Table 1: REST in SCLC vs. REST in Alzheimer's disease: the structural inversion

Context	**REST role**	**Geometric interpretation**
SCLC (non-NE subtype)	Hub: rises to SILENCE neuronal identity; pushes cells to SCLC-Y false attractor	REST is the disease-maintenance mechanism; inhibiting REST would restore neuronal identity
Healthy aging neuron	Guardian: rises to PROTECT the correct neuronal attractor by silencing cell-death, stress, and cell-cycle genes	REST rising = protective; loss of REST removes the checkpoint; REST is NOT the Hub
Alzheimer's neuron	Checkpoint LOST: tau aggregates sequester REST in autophagosomes; nuclear REST depleted; checkpoint removed	Loss of REST is a CONSEQUENCE of Hub activity (DYRK1A driving tau aggregation), not the Hub itself

The framework therefore correctly identifies DYRK1A — not REST — as the true Convergence Hub in Alzheimer's disease. REST loss is the mechanism by which DYRK1A drives the neuron to the false attractor (via tau-mediated REST sequestration), but REST loss is downstream of DYRK1A overactivation. The drug target is DYRK1A. Restoring REST by other means (independent of tau clearance) would address only the checkpoint, not the Hub, and would therefore be insufficient for durable correction.

3. Drift Event 1: Cellular Neuronal Identity Loss

3.1. The Identity Anchor: MEF2C and NEUROD2

Cellular Identity Anchor — Drift Event 1

MEF2C (Myocyte Enhancer Factor 2C):
Terminal selector transcription factor for neuronal identity maintenance in the cortex and hippocampus. Required not only to *establish* neuronal identity during development but to *maintain* it throughout adult life. Loss of MEF2C produces: impaired mitochondrial function in neurons, elevated oxidative stress, neurodegenerative phenotype in animal models, and shift of microglia to the hyperinflammatory/senescent state. MEF2C variants are genome-wide significant AD risk loci [3, 4]. Loss of MEF2C is upstream of Alzheimer's pathology, not downstream.

NEUROD2 (Neuronal Differentiation Factor 2):
Pioneer transcription factor of cortical excitatory neuron identity. Falls in AD brains by single-nucleus RNA-seq analysis. Falls at the same chromatin loci where H3K27me3 is anomalously lost. The correlation with DYRK1A overactivity is direct.

The neuronal Identity Axis ratio for Drift Event 1:

$$R_{\mathrm{AD}} = \frac{\text{MEF2C expression / NEUROD2 activity}}{\text{DYRK1A expression / tau phosphorylation load}} \tag{1}$$

3.2. The Convergence Hub: DYRK1A

DYRK1A (Dual-specificity tyrosine-Phosphorylation Regulated Kinase 1A) is the primary Convergence Hub for neuronal identity loss in Alzheimer's disease. It is the maintenance mechanism of the false attractor.

DYRK1A as Hub: the convergent mechanisms

DYRK1A phosphorylates **MEF2C** at Ser408 → MEF2C is inactivated → Identity Anchor suppressed → neuronal identity programme begins to drift.

DYRK1A phosphorylates **tau** at multiple sites (Thr181, Ser202, Thr231, Ser396, Ser404) → tau becomes hyperphosphorylated and aggregation-prone → tau aggregates form neurofibrillary tangles → tau aggregates sequester REST in autophagosomes → nuclear REST depleted → checkpoint removed.

DYRK1A promotes **aberrant cell cycle re-entry** via phosphorylation of LIN52 (MuvB complex) and CDK4/Rb pathway activation → post-mitotic neuron attempts to divide → abortive mitosis → apoptosis.

DYRK1A is overexpressed on chromosome 21 → in trisomy 21 (Down syndrome), constitutive DYRK1A overexpression → R_{AD} depressed from birth → AD onset at median age 40 in Down syndrome individuals. *This is the geometric derivation of the Down syndrome/Alzheimer's connection, derived from the Hub identity of DYRK1A.*

3.3. The Secondary Hub: Pathological Tau as Propagation Mechanism

The tau aggregate serves a dual role in the geometry: it is both a *product* of DYRK1A Hub activity (DYRK1A phosphorylates tau, driving aggregation) and a *spreading mechanism* for the Hub's effect (tau aggregates spread trans-synaptically, confirmed by prion-like propagation studies [8]).

The propagation cascade:

$$\underbrace{\text{DYRK1A}\uparrow}_{\text{Hub activated}} \longrightarrow \underbrace{\text{tau-p}\uparrow}_{\text{phosphorylation}} \longrightarrow \underbrace{\text{tau aggregates}}_{\text{NFT formation}} \longrightarrow \underbrace{\text{REST sequestered}}_{\text{checkpoint lost}} \longrightarrow$$

$$\longrightarrow \underbrace{\text{cell cycle entry}}_{\text{CDK4/Rb}} \longrightarrow \underbrace{\text{apoptosis}}_{\text{neuronal death}} \longrightarrow \underbrace{\text{tau released}}_{\text{trans-synaptic}} \longrightarrow \underbrace{\text{adjacent neuron}}_{\text{REST sequestered}} \longrightarrow \cdots$$

Each neuron receiving tau aggregates loses its nuclear REST, becomes vulnerable to DYRK1A-driven cell cycle re-entry, and dies — releasing more tau to the next neuron. The Braak staging of AD corresponds directly to this geometric propagation: the wave front of the false attractor spreading through the hippocampal circuit, the entorhinal cortex, and the neocortex [9].

3.4. The Sub-Type Structure and Cognitive Reserve

The R_{AD} framework predicts a continuous subtype structure, not a binary healthy/diseased classification:

Corollary 1 (Cognitive Reserve as High R_{AD})**.** *The phenomenon of "cognitive reserve" — individuals with confirmed amyloid-PET pathology who remain cognitively intact for years longer than expected — is predicted to be the population with measurably higher baseline R_{AD}. High MEF2C expression in hippocampal and entorhinal neurons sustains R_{AD} above threshold despite amyloid accumulation, preventing the DYRK1A overactivation cascade. The cognitive reserve population will show measurably higher MEF2C, lower DYRK1A phosphorylation of tau, and higher nuclear REST in accessible biomarker tissue compared to age-matched individuals with the same amyloid burden and earlier clinical symptoms.*

3.5. The Drug Target: DYRK1A Inhibition

The Hub identification ($\mathcal{H} = \text{DYRK1A}$) directly yields the primary drug target:

Drug Target: CNS-Penetrant DYRK1A Inhibitor

Target: DYRK1A kinase activity inhibition in cortical and hippocampal neurons.

Mechanism: DYRK1A inhibition → MEF2C phosphorylation reduced → MEF2C reactivated → neuronal identity programme restored. Simultaneously: tau phosphorylation at DYRK1A sites reduced → tau aggregation slowed → REST sequestration slowed → REST checkpoint partially restored.

Clinical candidates:

- **Leucettine L41**: DYRK1A inhibitor, Phase 2 clinical trial in Down syndrome/AD prevention [11]. Confirmed CNS penetrance. Reduces tau phos-

Table 2: R_{AD} sub-type structure and clinical predictions

R_{AD} level	Molecular state	Clinical phenotype	Framework prediction
High (healthy)	MEF2C high, DYRK1A low, REST nuclear, H3K27me3 preserved, tau unmethylated	Cognitively normal, no clinical symptoms even if amyloid-PET positive	Correct attractor. AD pathology present but not progressing. "Cognitive reserve."
Intermediate	MEF2C decreasing, DYRK1A rising, REST partially nuclear, early tau-p	MCI (Mild Cognitive Impairment); early memory complaints	Near-threshold. Both drifts beginning. Optimal intervention window.
Low	MEF2C low, DYRK1A dominant, REST depleted, tau aggregates spreading	Established AD. Progressive cognitive decline. Hippocampal volume loss.	False attractor. Neuronal death occurring. Drift Event 2 also active.
Floor	MEF2C absent, DYRK1A fully active, most hippocampal neurons dead	Late-stage AD. Loss of ADLs, language, personality.	Irreversible substrate loss. Drift Event 1 uncorrectable. Drift Event 2 intervention limited by surviving engram cell count.

phorylation and restores synaptic plasticity in animal models.

- **GNF2133**: DYRK1A inhibitor, preclinical AD models [12]. Shows tau reduction and cognitive improvement (Morris water maze) in 3xTg-AD mice.
- **EHT 1610**: Selective DYRK1A/CLK1 inhibitor, neuroprotective in AD cellular models.

Framework convergence: The framework derives DYRK1A as Hub from geometry. The field arrived at DYRK1A empirically (Down syndrome/AD connection, tau phosphorylation mapping). They converge on the same target. This is the same pattern as every other table entry.

4. Drift Event 2: Memory Identity Loss

4.1. Memory as an Identity in Waddington Space

Drift Event 2 is the framework's most novel contribution to the AD derivation. It extends the concept of cellular identity to a second level: the identity of what a cell *holds*.

What Memory Identity Means Geometrically

A memory is encoded as a specific pattern of: (1) long-term synaptic potentiation (LTP) at particular dendritic spines of the engram cell, (2) epigenetic marks (H3K27ac, H3K4me3) at the gene loci whose products maintain those synaptic weights (Arc, Fos, Homer1a, GluR1/2), (3) CREB-mediated transcriptional programme that sustains the synaptic signature over time.

Together, these constitute an *attractor state* for the engram cell at the *informational level*: a stable configuration that encodes a specific memory. The memory is accessible so long as the attractor is stable. The memory becomes inaccessible — but not erased — when the attractor drifts below retrieval threshold. The identity of *what the cell holds* has drifted, not the identity of *what the cell is*.

4.2. The Memory Identity Anchor: CREB/BDNF/TrkB

$$\mathcal{A}_{\text{mem}} = \text{CREB}^{(\text{phospho-Ser133})} \cdot \text{BDNF} \cdot \text{TrkB} \tag{2}$$

- **CREB (Ser133):** Phospho-CREB is the master activator of memory-gene transcription. CREB phosphorylation at Ser133 drives expression of BDNF, Arc, Fos, Homer1a, and other LTP maintenance genes. Loss of CREB-Ser133 activity means the synaptic weights at engram cell dendritic spines are not being actively maintained. The synaptic signature erodes. The memory becomes inaccessible [10, 19].
- **BDNF:** Brain-Derived Neurotrophic Factor. Acts via TrkB to sustain dendritic spine density, LTP maintenance, and CREB transcriptional activity in hippocampal neurons. BDNF falls early in AD (confirmed: entorhinal cortex BDNF reduction precedes clinical symptoms) [13].
- **TrkB:** BDNF receptor. TrkB signalling activates CaMKIV, which phosphorylates CREB, completing the memory maintenance loop. Loss of TrkB density is directly associated with synaptic loss in AD [14].

4.3. The Memory Identity Hub: HDAC2/HDAC3

$$\mathcal{H}_{\text{mem}} = \text{HDAC2} \cdot \text{HDAC3} \tag{3}$$

HDAC2 and HDAC3 deacetylate H3K27ac at memory-encoding loci (Arc, Fos, GluR1/2, Homer1a, BDNF). Deacetylation closes the chromatin at these loci → gene expression falls → synaptic maintenance proteins are no longer produced → LTP decays → the memory's epigenetic signature erodes toward baseline (inaccessible state).

- **HDAC2** is the primary identity axis component: elevated in AD hippocampus [15]; overexpression produces memory impairment in mice; specific knockout of HDAC2 restores memory access in AD mouse models [16]. HDAC2 is the molecular lock on the memory attractor.
- **HDAC3** co-participates in the deacetylation of the same loci, particularly in the consolidation phase of new memory formation. HDAC3 inhibition enhances long-term memory formation in rodents [17].

The memory identity axis ratio:

$$R_{\text{mem}} = \frac{\text{CREB-Ser133 activity} \cdot \text{BDNF} \cdot \text{TrkB}}{\text{HDAC2} \cdot \text{HDAC3}} \tag{4}$$

When R_{mem} falls below retrieval threshold: the memory becomes inaccessible. The information is preserved in the dormant synaptic architecture but cannot be retrieved because the epigenetic signature has drifted below expression threshold. This is *not* erasure. It is *inaccessibility*.

4.4. The Information Preservation Hypothesis

The most important structural claim of the two-drift model: In early and middle Alzheimer's disease, before the engram neurons themselves die (Drift Event 1 reaches the engram cell), the memories are inaccessible, not erased. The synaptic architecture that encodes the memory is degraded but not fully destroyed. The epigenetic marks at memory-encoding loci have drifted below retrieval threshold but the loci still exist. Restoring R_{mem} above retrieval threshold — by HDAC2/3 inhibition, TrkB agonism, or CREB phosphorylation enhancement — may restore access to memories that appeared clinically "lost."

Experimental confirmation (partial): Optogenetic activation of engram cells in AD mouse models (APP/PS1) that can no longer retrieve memories behaviorally *restores* memory recall during the optogenetic stimulation [20]. The information was there. It was inaccessible. The optogenetic stimulation bypassed the failed retrieval mechanism. This is the in vivo proof of the Information Preservation Hypothesis.

4.5. The Memory Identity Drug Targets

Drug Targets for Drift Event 2: Memory Identity Restoration

Target 1: HDAC2 selective inhibitor.
Inhibit HDAC2 specifically at memory-encoding chromatin loci → H3K27ac preserved → Arc, Fos, BDNF, GluR1/2 expression maintained → synaptic maintenance proteins produced → LTP sustained → R_{mem} raised.

Candidate: Selective HDAC2 inhibitor (not pan-HDAC — pan-HDAC inhibitors such as vorinostat have cardiovascular toxicity profiles that preclude chronic use in AD population). Current state: selective HDAC2 inhibitors in medicinal chemistry development. No clinical candidate yet approved for AD specifically.

Target 2: TrkB agonist.
Activate TrkB directly (bypassing BDNF, which does not adequately cross the blood-brain barrier at therapeutic doses) → CaMKIV/CREB phosphorylation loop restored → BDNF gene expression re-initiated → R_{mem} raised.

Candidate: ANA 12 (TrkB antagonist — note: this is the WRONG direction; what is needed is TrkB agonist). 7,8-DHF (7,8-Dihydroxyflavone): confirmed TrkB partial agonist, crosses the BBB, improves cognition in 3xTg-AD mice [21]. LM22A-4: TrkB agonist, preclinical AD neuroprotection [22].

Target 3: CREB activator.
Directly enhance CREB-Ser133 phosphorylation via PDE4 inhibition (raises cAMP → PKA → CREB-Ser133) or direct PKA activator → CREB transcriptional programme restored → memory maintenance gene expression reinstated.

Candidate: Rolipram (PDE4 inhibitor): improves memory in multiple AD models; tolerability issues (emesis) in humans but newer selective PDE4B inhibitors are in development.

5. The Down Syndrome / Alzheimer's Connection: A Geometric Derivation

The connection between trisomy 21 (Down syndrome) and early-onset Alzheimer's disease is the single clearest external validation of the DYRK1A Hub identification.

The geometric derivation:
DYRK1A maps to chromosome 21 (21q22.13). Trisomy 21 produces three copies of DYRK1A instead of two. Constitutive 1.5× overexpression of DYRK1A from birth. From the framework:

$$R_{\mathrm{AD}} = \frac{\mathrm{MEF2C}}{\mathrm{DYRK1A}}$$

If DYRK1A is constitutively elevated from birth at 1.5× normal, R_{AD} is constitutively depressed by ≈33% below the normal baseline for the entire lifetime. The neuron spends its entire life closer to the threshold. Every additional perturbation (aging, inflammation, stress) that would normally be tolerated without crossing the threshold crosses it decades earlier in a DYRK1A-overexpressing neuron.

Clinical prediction confirmed: Virtually all individuals with Down syndrome develop Alzheimer's-type neuropathology by age 40 and clinical dementia by age 50 [23]. This is the constitutive R_{AD} depression predicted by the geometry. The Hub is overexpressed from conception. The false attractor is therefore reached decades earlier.

Treatment implication: In Down syndrome, the Hub overexpression is constitutional and cannot be reversed by a kinase inhibitor indefinitely. The correct intervention is lifelong DYRK1A inhibitor prophylaxis beginning in early adulthood (before the threshold crossing), not AD treatment after onset. Leucettine L41 Phase 2 trial is running in this population. This is the correct indication derived from the geometry.

6. The Integrated Treatment Protocol

6.1. Architecture: Two Drift Events, Two Target Sets, One Sequence

The integrated protocol addresses both drift events with distinct but temporally overlapping interventions. The sequence is determined by the geometric dependencies.

Table 3: Integrated treatment protocol by disease stage

Stage	Drift Event 1 intervention	Drift Event 2 intervention	Primary goal
Pre-symptomatic (high amyloid-PET, normal cognition)	DYRK1A inhibitor (Leucettine L41) prophylactic	HDAC2i + TrkB agonist prophylactic	Prevent threshold crossing on both drift axes simultaneously
MCI (early)	DYRK1A inhibitor (full therapeutic dose) + MEF2C agonism support	HDAC2i + TrkB agonist + PDE4B inhibitor (CREB activation)	Halt neuronal death; restore memory identity for accessible memories; rescue recently inaccessible memories
Established AD (moderate)	DYRK1A inhibitor (slows further loss); TrkB agonist (neuroprotection)	HDAC2i (restore access to surviving engram cells); intensive cholinergic support	Slow Drift Event 1 progression; maximise recovery of inaccessible memories in surviving neurons
Late-stage AD	Neuroprotective support; tau clearance (lecanemab adjunct)	HDAC2i where surviving engram substrate remains	Quality of life; slow further irreversible loss

6.2. The Critical Sequence Rule

The sequence is not arbitrary.

Drift Event 2 intervention (HDAC2 inhibition) MUST be applied while the engram neurons are still alive. HDAC2 inhibition at dead neurons recovers nothing. This is why the pre-symptomatic and MCI intervention windows are geometrically critical: they are the windows in which the engram cell substrate that holds the inaccessible memories still exists. Every month of delayed Drift Event 1 treatment (DYRK1A inhibition) is months of engram cells dying — shrinking the substrate pool available for Drift Event 2 recovery.

6.3. Amyloid Therapy Repositioned

The framework repositions amyloid-targeting therapies (lecanemab, donanemab) within the geometry:

Amyloid-β accumulation is *upstream* of the Hub cascade (Aβ oligomers activate multiple

kinases including DYRK1A and GSK3β, which together hyperphosphorylate tau). Amyloid clearance is therefore a partial upstream intervention: it reduces one of the inputs to the Hub cascade without targeting the Hub itself.

Geometric placement of amyloid therapy:
Aβ oligomers $\rightarrow$ DYRK1A/GSK3β activation $\rightarrow$ tau phosphorylation $\rightarrow$ Hub cascade.

Lecanemab clears Aβ plaques $\rightarrow$ reduces oligomeric Aβ $\rightarrow$ reduces one activating input to the Hub $\rightarrow$ partial R_{AD} improvement.

This is why lecanemab shows modest but real clinical effect (29% slowing of decline in CDR-SB at 18 months [24]) but not halting or reversal: it reduces Hub activation pressure but does not address the Hub directly. DYRK1A inhibitor + lecanemab should produce additive effects because they target different nodes in the cascade.

7. Eight Falsifiable Predictions for the Record

All predictions are derived from the framework and timestamped 2026-03-09.

Prediction 1 (Cognitive reserve = high R_{AD})**.** *Individuals with confirmed amyloid-PET positivity and no cognitive symptoms ("resilient" population) will show measurably higher MEF2C expression in accessible biomarker tissue (CSF or iPSC-derived hippocampal neurons), lower DYRK1A-mediated tau phosphorylation (p-tau 181, p-tau 231 in CSF), and higher nuclear REST levels compared to age-matched individuals with the same amyloid burden and established MCI or AD symptoms. The cognitive reserve effect is a high-*R_{AD} *effect.*

Prediction 2 (DYRK1A inhibitor + lecanemab additivity)**.** *In a clinical trial combining a CNS-penetrant DYRK1A inhibitor (Leucettine L41 or equivalent) with lecanemab in early AD, the combination arm will show statistically significantly greater slowing of CDR-SB decline than either monotherapy arm, because the combination addresses both the upstream amyloid input (lecanemab) and the Hub itself (DYRK1A inhibitor). The additive effect will be largest in the MCI population where both drugs can reach enough surviving neurons to act on.*

Prediction 3 (HDAC2 inhibition rescues optogenetically inaccessible memories)**.** *In an AD mouse model (APP/PS1 or 3xTg-AD) at the stage where specific memories are not behaviourally accessible, HDAC2-selective inhibition will restore memory retrieval on the fear-conditioning or object-recognition task at a rate statistically greater than vehicle control, without requiring optogenetic stimulation of the engram cell. This is the drug-mediated equivalent of the Roy 2016 optogenetic result [20], achieved pharmacologically by raising* R_{mem} *above retrieval threshold.*

Prediction 4 (7,8-DHF + DYRK1A inhibitor synergy in 3xTg-AD)**.** *In 3xTg-AD mice, combined treatment with 7,8-DHF (TrkB agonist, Drift Event 2 intervention) and GNF2133 or EHT 1610 (DYRK1A inhibitor, Drift Event 1 intervention) will show greater improvement in both synaptic density (Drift Event 2 biomarker) and tau phosphorylation (Drift Event 1 biomarker) than either drug alone, consistent with the two-drift model predicting*

independent but complementary mechanisms.

Prediction 5 (R_{AD} CSF biomarker predicts conversion rate)**.** *A composite CSF biomarker panel of MEF2C/DYRK1A ratio (or equivalent proxy: p-tau 231 as DYRK1A activity surrogate; NEUROD2 as MEF2C activity surrogate) will predict conversion from cognitively normal amyloid-positive to MCI within 3 years with greater accuracy than amyloid or tau burden alone.*

Prediction 6 (Down syndrome prophylactic window for Leucettine)**.** *Leucettine L41 initiated in Down syndrome individuals between ages 25 and 35 (before the median threshold crossing) will significantly reduce CSF tau phosphorylation markers and delay Alzheimer's symptoms onset by $\geq$5 years compared to untreated Down syndrome controls, because it directly addresses the constitutional DYRK1A overexpression that is the geometric cause of early R_{AD} threshold crossing in this population.*

Prediction 7 (Selective HDAC2 inhibitor outperforms pan-HDAC in AD memory models)**.** *A selective HDAC2 inhibitor with confirmed specificity over HDAC1 and HDAC3 will show equivalent memory restoration in an AD behavioural model compared to non-selective HDAC inhibitors, with significantly reduced haematological and cardiovascular adverse effects, confirming the geometric prediction that HDAC2 (not HDAC1 or the class I/II panel) is the specific Hub at memory-encoding chromatin loci.*

Prediction 8 (REST restoration insufficient as monotherapy)**.** *Restoration of nuclear REST levels (by inhibiting the tau-mediated autophagosomal sequestration of REST, for example via TFEB activation increasing autophagosomal clearance) without simultaneous DYRK1A inhibition will not prevent progression of tau phosphorylation or neuronal identity loss in an AD model, because REST restoration addresses the lost checkpoint but not the active Hub (DYRK1A) that is continuing to phosphorylate MEF2C and tau independently of REST status.*

8. Falsifiability

Following the governing framework principle [2]:

8.1. Type B Failure (framework-invalidating)

If a CNS-penetrant DYRK1A inhibitor administered to confirmed high-DYRK1A AD neurons (validated by CSF/iPSC biomarker panel) produces: (1) confirmed DYRK1A activity reduction AND (2) confirmed partial MEF2C reactivation (by ChIP-seq or iPSC functional assay) AND (3) no measurable change in tau phosphorylation rate, synaptic density, or cognitive decline — then DYRK1A is not the functional Hub (it is inactivated but the disease progresses from another driver entirely) and the Identity Anchor identification (MEF2C) does not predict neuronal survival. The framework application to AD is invalidated; Hub and Anchor must be re-derived.

8.2. Type A Failures (informative, not invalidating)

- *DYRK1A inhibitor halts neuronal loss but memory impairment persists:* Structural explanation: Drift Event 2 (memory identity) is mechanistically independent of Drift

Event 1. Halting neuronal death does not restore the eroded synaptic signature of inaccessible memories. Both drift events must be treated. HDAC2 inhibitor must be added. The framework predicts this outcome in any trial that targets only one drift.

- *HDAC2 inhibitor restores memory access but does not prevent further neuronal loss:* Structural explanation: Correct. HDAC2 inhibition raises R_{mem} (memory identity) but does not address R_{AD} (neuronal identity). This is the predicted single-drift result. A positive result here confirms the two-drift model, not the adequacy of single-target therapy.
- *HDAC2 inhibition is ineffective in late-stage AD:* Structural explanation: In late-stage AD, the engram neurons that held the inaccessible memories have died (Drift Event 1 reached them). There is no substrate for HDAC2 inhibition to act on. This is Stage-dependent failure. The framework predicts it. The prediction narrows the effective intervention window to early/middle stage. It does not invalidate the mechanism.

9. Discussion

9.1. Why Alzheimer's Has Resisted Disease Modification for 30 Years

The amyloid hypothesis has dominated AD research for three decades. The geometric framework does not reject the amyloid hypothesis — it repositions it. Amyloid is an upstream activator of the Hub cascade, not the Hub itself. Clearing amyloid (lecanemab, donanemab) reduces Hub pressure but leaves the Hub (DYRK1A) active through other inputs (aging, inflammation, oxidative stress, tau propagation). The clinical effect is modest and never reversal because the Hub is not addressed. The same error — treating a consequence of the Hub rather than the Hub itself — accounts for the failure of tau clearance antibodies as monotherapy: tau clearance removes a propagation mechanism for the Hub effect, but DYRK1A continues to phosphorylate new tau even as the old aggregates are cleared.

The framework predicts that the combination of upstream amyloid clearance (lecanemab) + Hub inhibition (DYRK1A inhibitor) + Identity Anchor restoration (MEF2C agonism, TrkB activation) + memory identity restoration (HDAC2 inhibition) will produce the first genuinely disease-modifying, symptom-reversing protocol in AD. No trial has yet combined these four components. The geometry derives why they must be combined.

9.2. The Inaccessible Memory Hypothesis and Therapeutic Implications

The most clinically significant claim in this document is the Information Preservation Hypothesis: that in early-to-middle Alzheimer's disease, many "lost" memories are not lost but inaccessible — and that pharmacologically raising R_{mem} above retrieval threshold could restore access to those memories in the window before their substrate neurons die.

The Roy 2016 optogenetic experiment [20] already demonstrates this in mice. What is needed is the drug equivalent of the optogenetic intervention: HDAC2 inhibition that restores the epigenetic signature of the memory-encoding chromatin without global genomic

disruption.

If this holds in humans, it implies that the most urgent clinical objective in early AD is not merely slowing decline but actively recovering already-lost cognitive function, which is a qualitatively different goal from all current AD therapeutics.

9.3. On Derivation Origin

This derivation was produced by an independent researcher with no medical background, no institutional affiliation, and no access to primary data. The two-drift model was derived from the geometry of the Identity Axis Framework applied to what neurons do (maintain identity over a lifetime) and what memories are (informational attractors encoded in synaptic/epigenetic signatures). The clinical implication — that there exists a therapeutic window in which inaccessible memories could be recovered — was a geometrically derived prediction, not an empirical observation. The Roy 2016 optogenetic result was found afterward as a confirmation. This is the framework's predictive mode: geometry first; literature confirms.

10. Derivation Record and Provenance

Zero structural contradictions. All framework derivations concerning Alzheimer's disease are consistent with the published literature. The stake established in the Falsifiability Theorem [2] stands.

11. Conclusion

Alzheimer's disease is two identity drift events, not one. The neuron is dying because its identity programme (MEF2C/NEUROD2) is being suppressed by an overactive Hub (DYRK1A) that simultaneously drives tau aggregation, removes the REST checkpoint, and forces aberrant cell cycle re-entry. The memory is becoming inaccessible because the informational structure of the engram — the epigenetic signature (H3K27ac at Arc, Fos, BDNF, GluR1/2 loci) that constitutes the memory — is being erased by an overactive memory Hub (HDAC2/HDAC3) while the memory identity anchor (CREB/BDNF/TrkB) falls.

These are the same geometric mechanism at two different scales. The Waddington attractor landscape applies not only to the identity of a cell but to the identity of what a cell holds. Both levels are derivable, measurable, and druggable.

The drug targets are derivable: DYRK1A inhibition for the cellular drift; HDAC2 inhibition + TrkB agonism for the memory drift. Neither target is new. Both are in clinical development. What is new is their geometric integration into a two-drift model with a defined treatment sequence, a defined intervention window (before engram neurons die), and a defined prediction (memories that appear lost may be recoverable in the early window by raising R_{mem} above retrieval threshold).

Table 4: Confirmation status as of 2026-03-09

Claim	Status	Source
MEF2C as neuronal Identity Anchor in AD	Confirmed	[3, 4]
NEUROD2 reduction in AD by snRNA-seq	Confirmed	AD Atlas 2024
DYRK1A as Hub (MEF2C phosphorylation)	Confirmed	[10]
REST as neurodegeneration checkpoint	Confirmed	[5, 6]
REST loss via tau sequestration	Confirmed	[7]
Tau prion-like propagation	Confirmed	[8]
DYRK1A inhibitor in trials (DS/AD)	Confirmed	[11]
HDAC2 elevated in AD hippocampus	Confirmed	[15]
HDAC2 KO restores memory in AD mice	Confirmed	[16]
BDNF early reduction in AD	Confirmed	[13]
7,8-DHF (TrkB agonist) in AD mice	Confirmed	[21]
Optogenetic engram recall in APP/PS1 mice	Confirmed	[20]
Down syndrome AD onset age 40–50	Confirmed	[23]
Lecanemab 29% CDR-SB slowing	Confirmed	[24]
Cognitive reserve = measurably higher R_{AD}	Predicted	Not yet tested
DYRK1A inhibitor + lecanemab additivity	Predicted	Not yet tested
HDAC2i restores pharmacological engram recall	Predicted	Not yet tested
R_{AD} CSF panel outperforms amyloid/tau alone as predictor	Predicted	Not yet validated
Leucettine prophylaxis delays AD in Down syndrome	Predicted	Trial ongoing
REST restoration insufficient without DYRK1A inhibitor	Predicted	Not yet tested
Selective HDAC2i over pan-HDAC for AD memory	Predicted	Not yet in trials
7,8-DHF + DYRK1A inhibitor synergy in 3xTg-AD	Predicted	Not yet tested

The Down syndrome connection is the geometric proof: constitutive DYRK1A overexpression from chromosome 21 trisomy produces early threshold crossing and AD onset by age 40 in virtually every affected individual. The Hub is identified. The gene is on chromosome 21. The consequence is exactly what the geometry predicts.

The geometry says the intervention window is open. Before the neurons die, the memories may still be there.

"I believe that is absolutely a matter of identity drift of some sort. It is a person's memory structure fading."

Both statements are geometrically correct. At two different levels simultaneously. Each with its own Hub. Each with its own key.

— Eric Robert Lawson, 2026-03-09

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Declarations

Competing interests: None.
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Data availability: No new data generated. All framework derivations at github.com/Eric-Robert-La
Ethics: Theoretical derivation. No patient data. No human subjects. This document does not constitute clinical advice of any kind.

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Appendix: Document Metadata

Field	Value
Document ID	ALZHEIMERS_ATTRACTOR_GEOMETRY
Type	Full principles-first derivation — Identity Axis Framework applied to Alzheimer's disease
Version	1.0
Date	2026-03-09
Status	ACTIVE — FORMAL RECORD
Author	Eric Robert Lawson / OrganismCore
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Repository	`attractor-oncology`
Framework DOI	10.5281/zenodo.18898788
License	CC BY 4.0
Framework class	Dual simultaneous identity drift (first entry of this class)
Drift Event 1 cell type	Cortical and hippocampal excitatory neurons and inhibitory interneurons
Drift Event 1 Anchor	MEF2C / NEUROD2
Drift Event 1 Hub	DYRK1A (primary); pathological tau (propagation)
Drift Event 1 drug target	CNS-penetrant DYRK1A inhibitor (Leucettine L41, GNF2133)
Drift Event 2 cell type	Engram cells (memory-encoding hippocampal and cortical neurons)
Drift Event 2 Anchor	CREB-Ser133 / BDNF / TrkB
Drift Event 2 Hub	HDAC2 / HDAC3
Drift Event 2 drug target	Selective HDAC2 inhibitor + TrkB agonist (7,8-DHF / LM22A-4) + PDE4B inhibitor (CREB activation)
Key novel claim	Information Preservation Hypothesis: "lost" memories in early AD are inaccessible, not erased; HDAC2 inhibition may restore access
Down syndrome connection	Constitutional DYRK1A overexpression (chr21 trisomy) = constitutively depressed R_{AD} from birth = AD onset 30–40 years earlier
Amyloid repositioning	Amyloid upstream of Hub (not the Hub itself); lecanemab is partial upstream input reduction

This document does not constitute clinical advice. It is a mathematical-geometric derivation from first principles. All clinical decisions must involve qualified medical professionals.